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(54) **HEAT-INTEGRATED TRANSFORMATIVE
CARBON DIOXIDE CAPTURE PROCESS**

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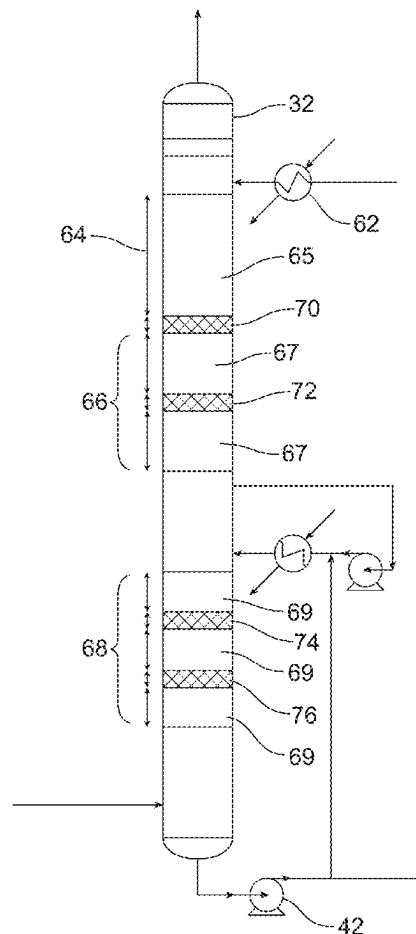
Related U.S. Application Data

(63) Continuation-in-part of application No. 17/883,482,
filed on Aug. 8, 2022, which is a continuation-in-part
of application No. 16/711,633, filed on Dec. 12, 2019,
now abandoned.

(60) Provisional application No. 62/779,702, filed on Dec.
14, 2018.

(57) **ABSTRACT**

An apparatus includes an absorber having a first packing section, a second packing section and a third packing section. The first packing segment includes a first structured packing, having a first specific surface area SA 1, the second packing segment includes a second structured packing, having a second specific surface area SA 2, and the third packing segment includes a third structured packing, having a third specific surface area SA 3 where SA 1<SA 2<SA 3. The structured packing in the various packing segment may be periodically interrupted with one or more layers of random packing.



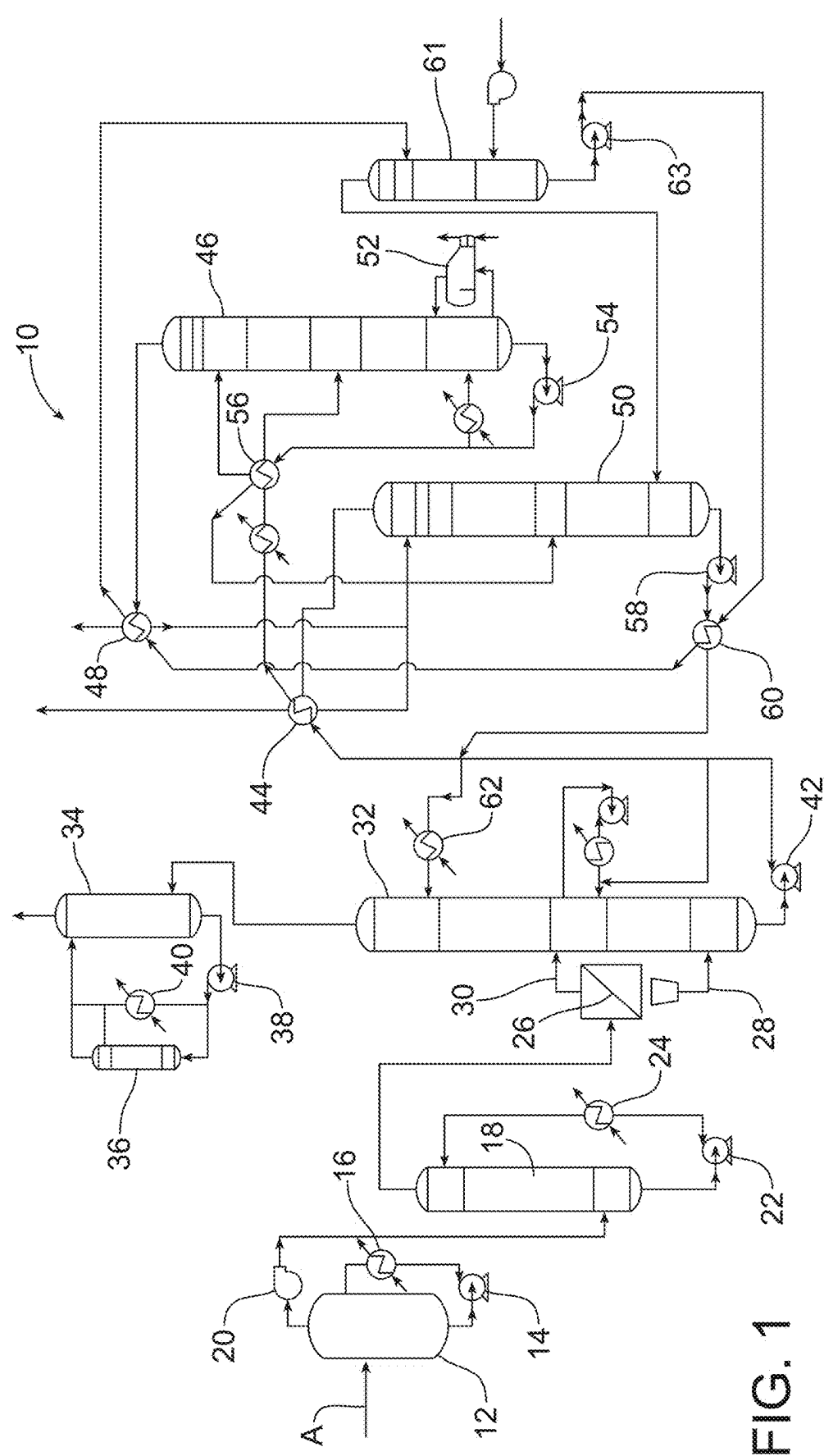


FIG. 1

FIG. 2

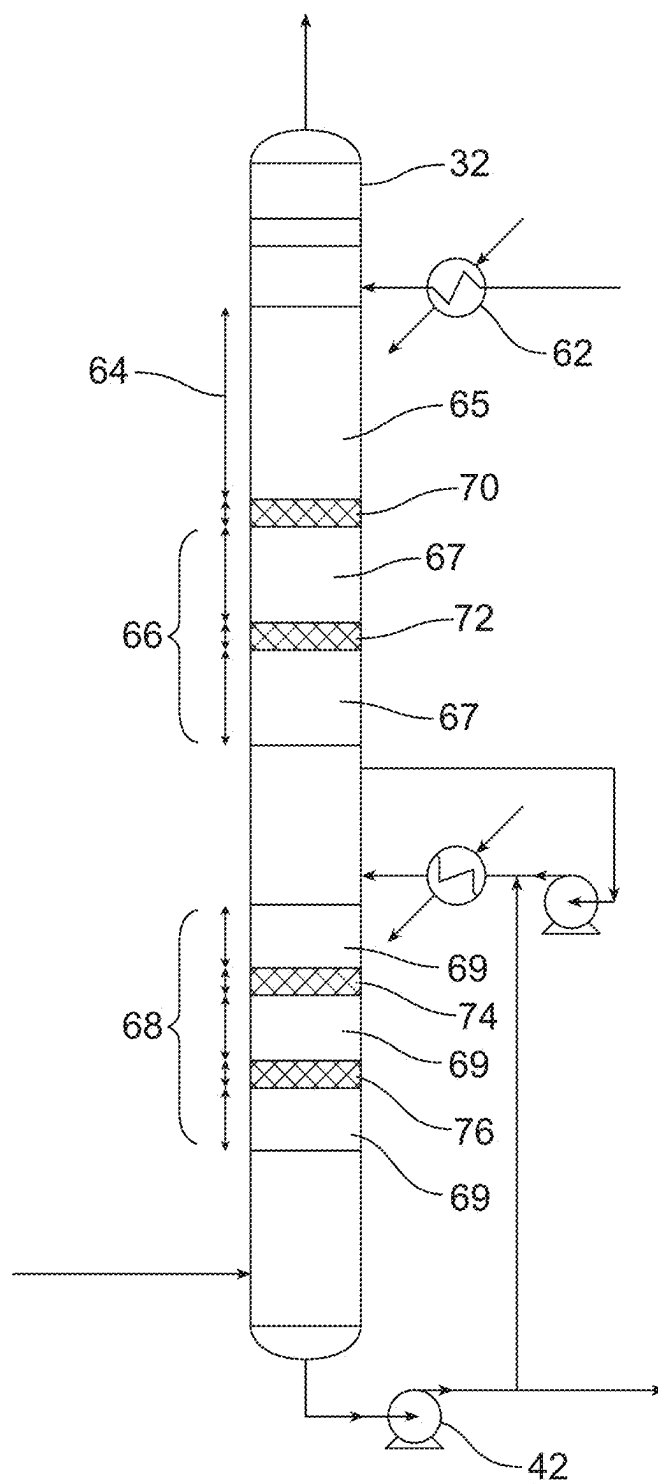


FIG. 3

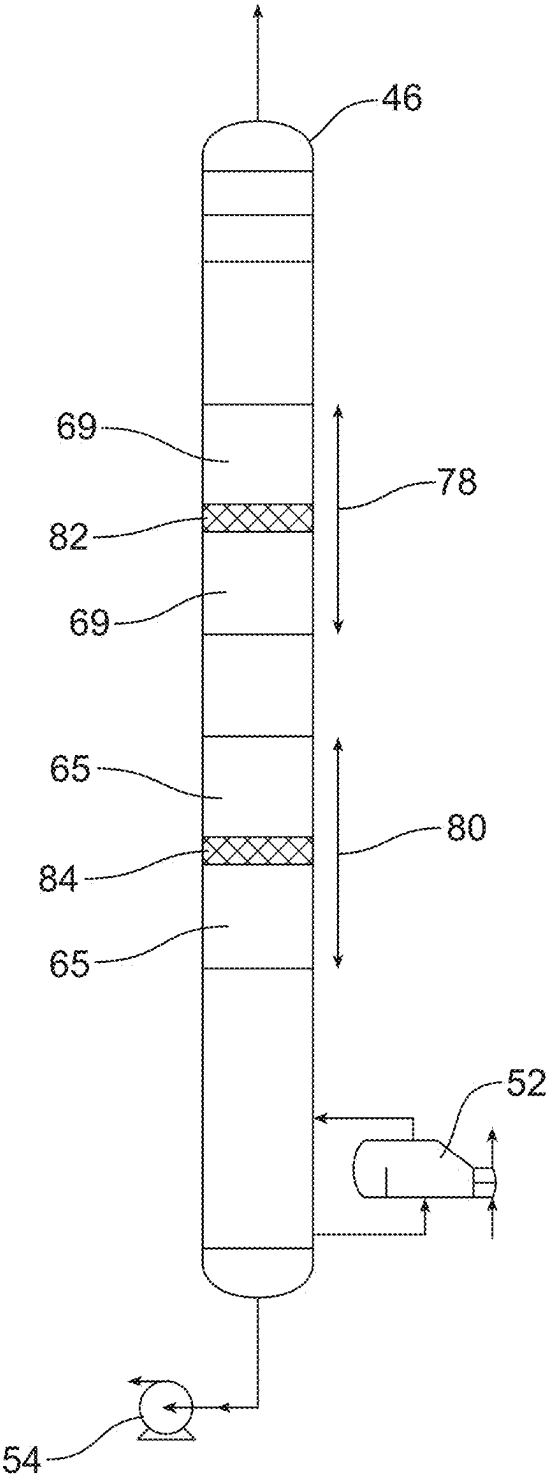


FIG. 4

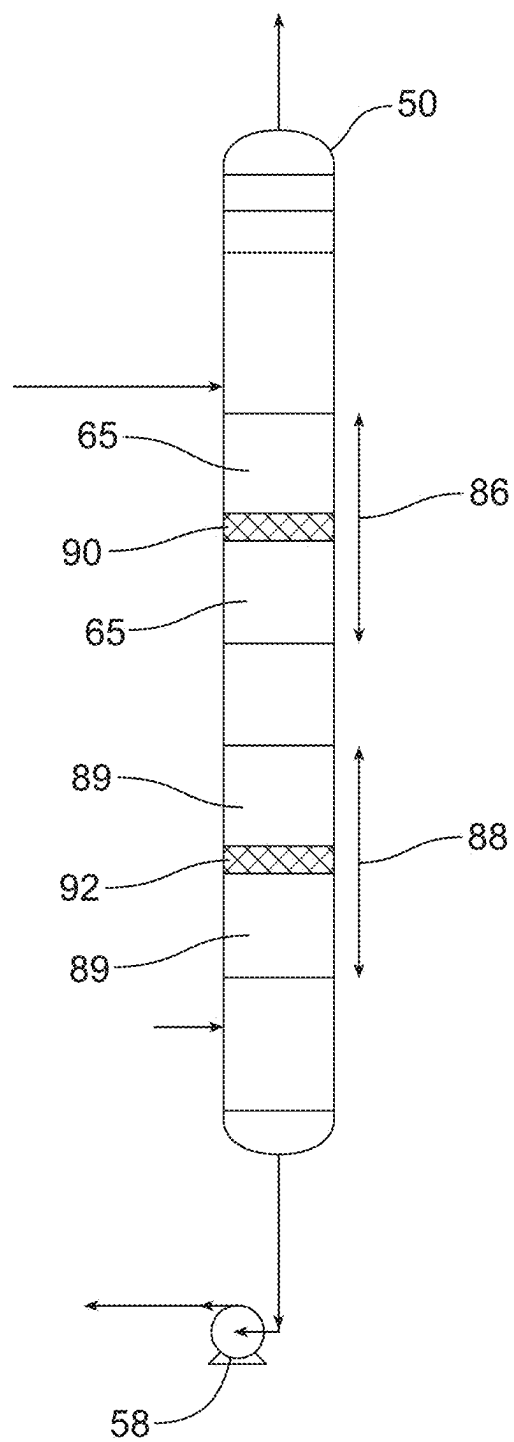


FIG. 5

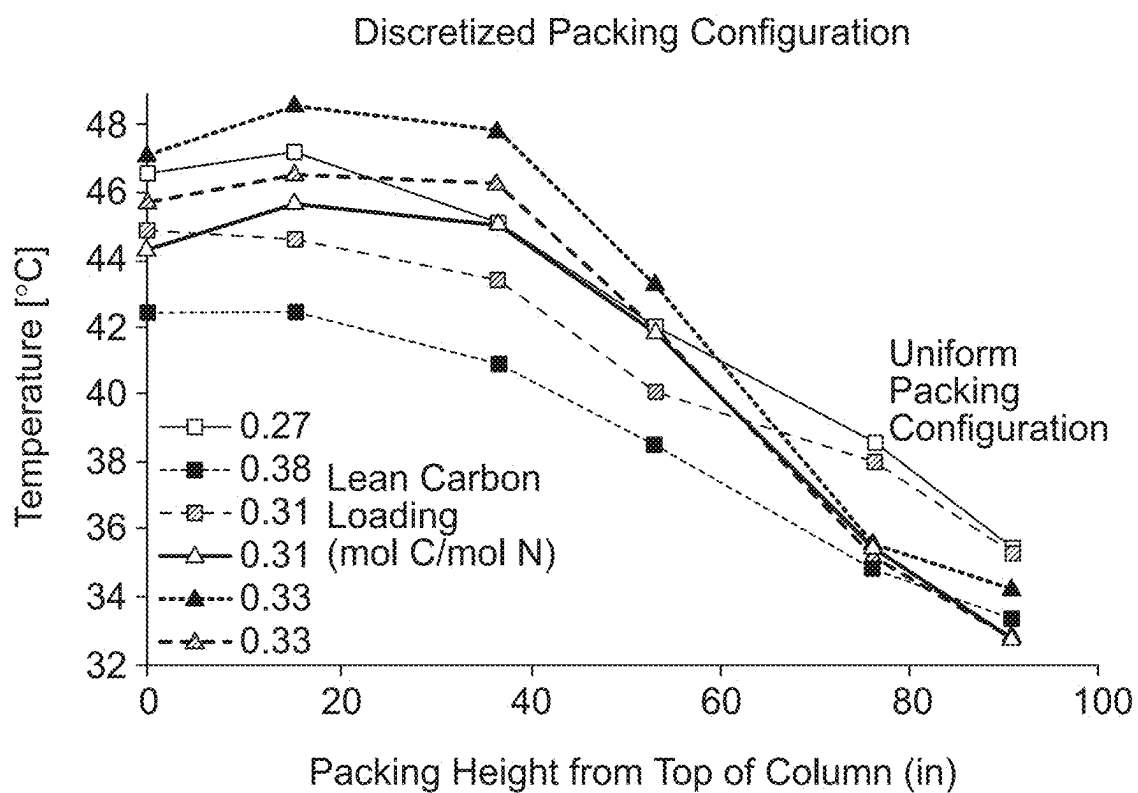


FIG. 6

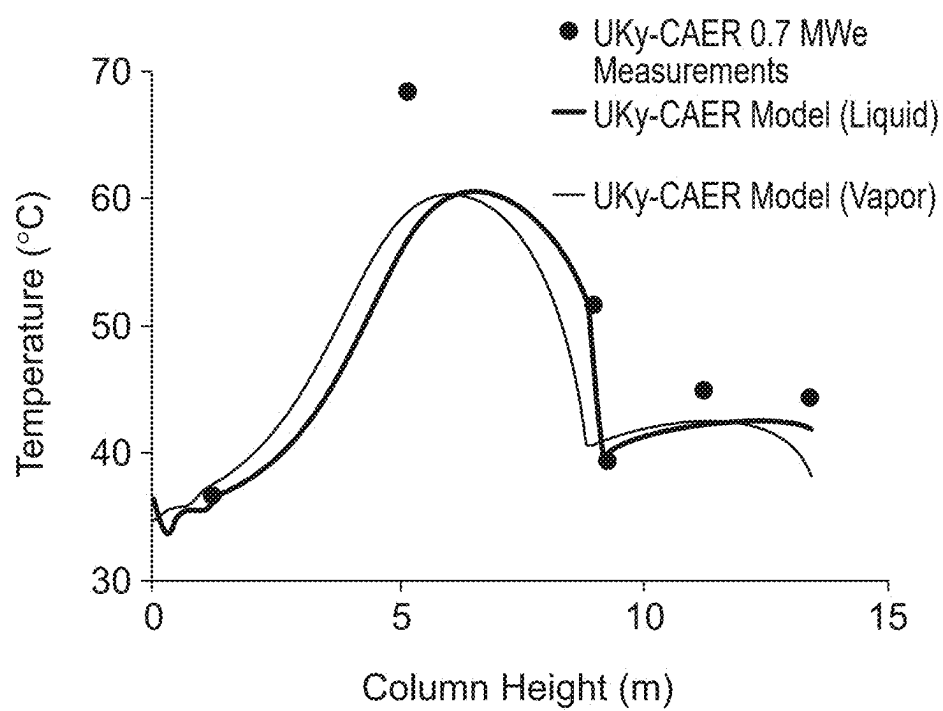


FIG. 7

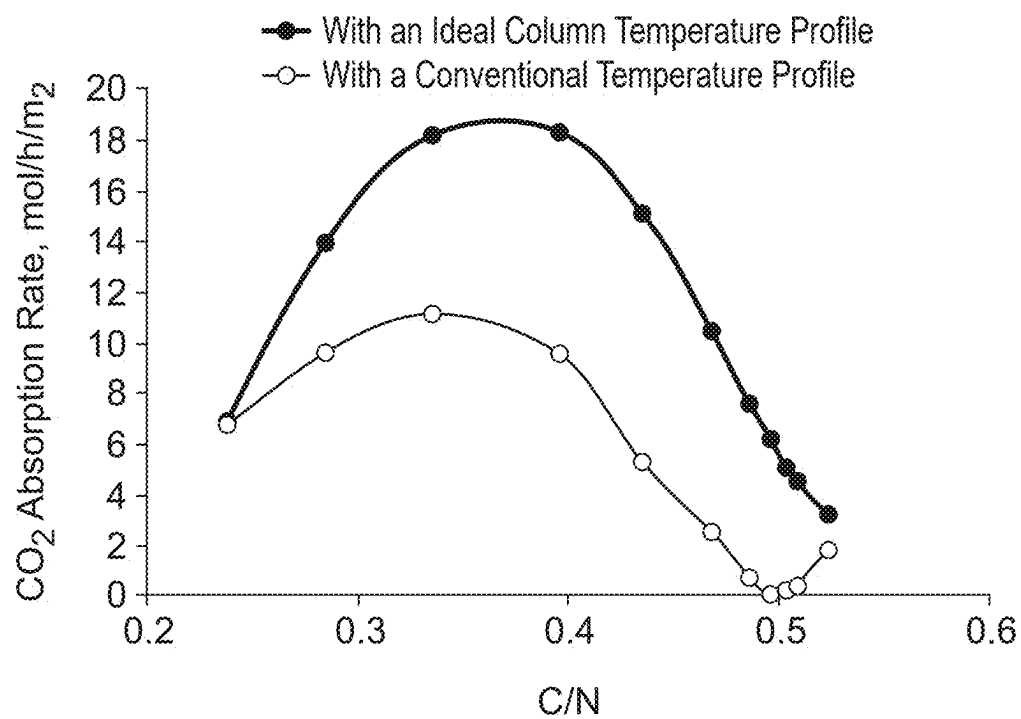


FIG. 8

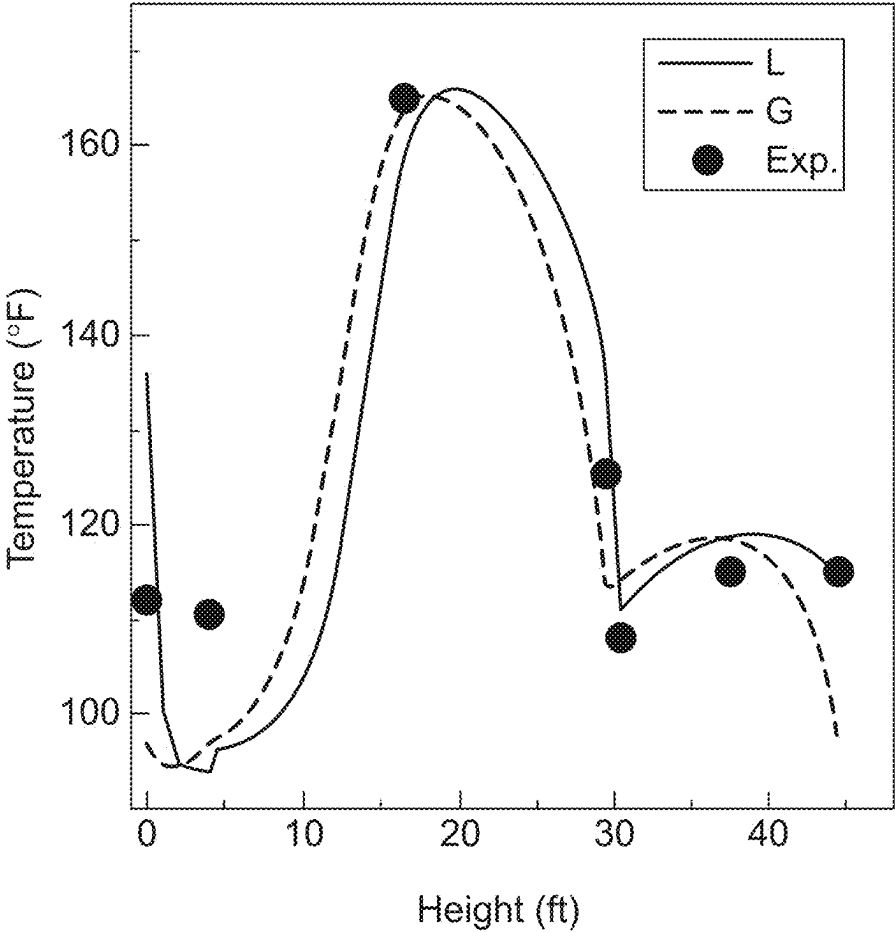


FIG. 9

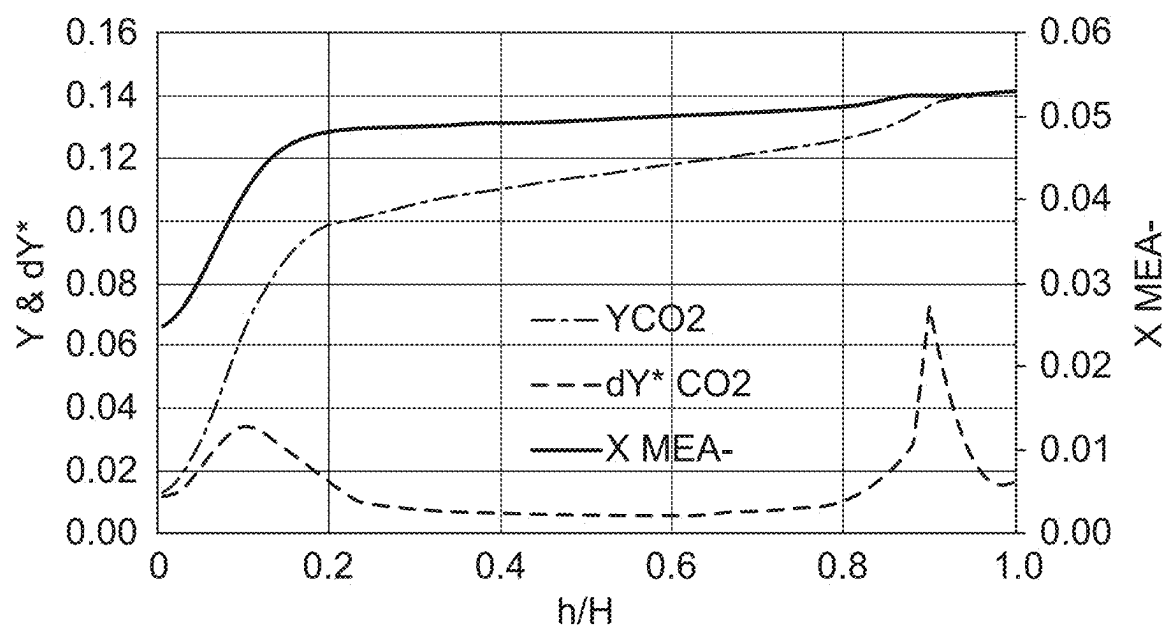
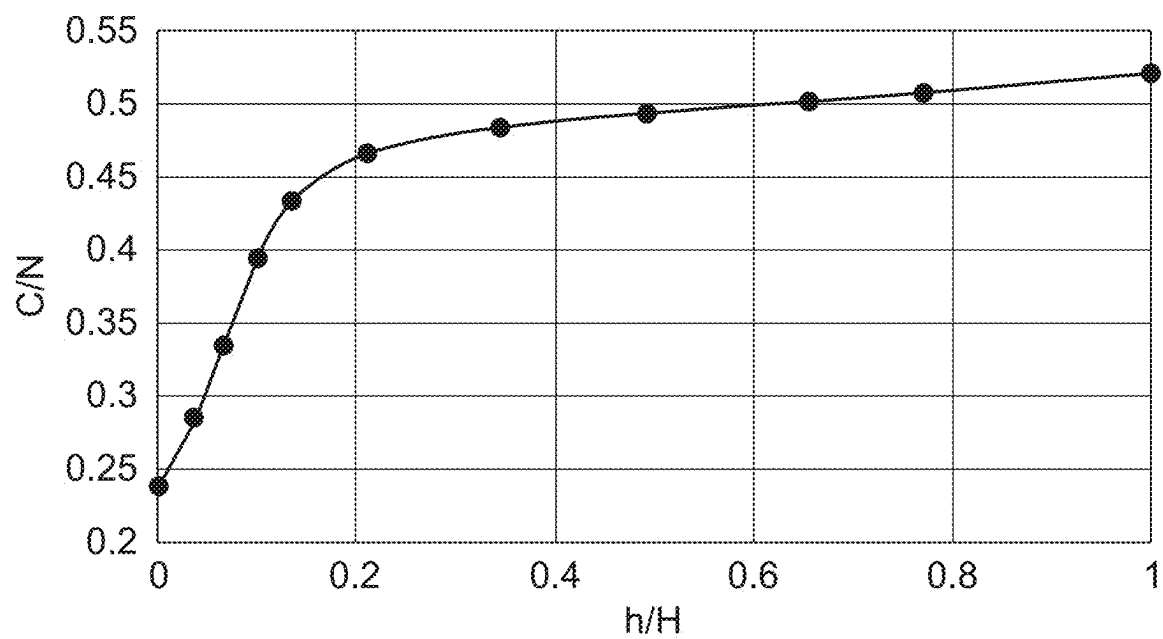
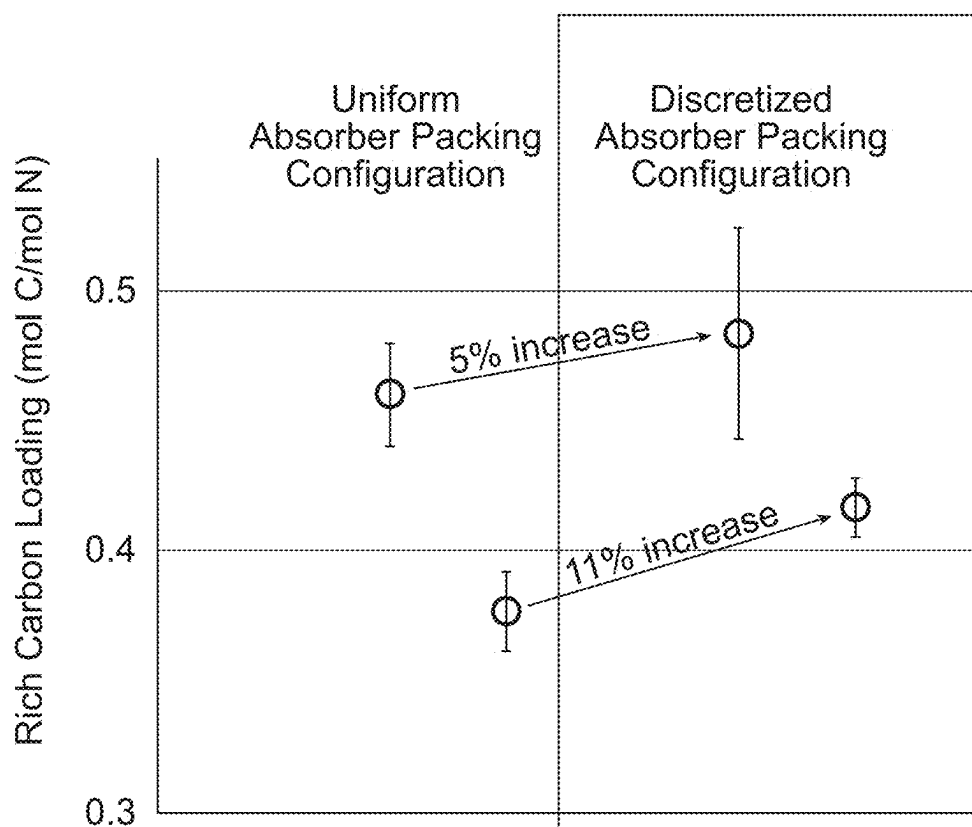


FIG. 10



HEAT-INTEGRATED TRANSFORMATIVE CARBON DIOXIDE CAPTURE PROCESS

RELATED APPLICATIONS

[0001] This application is a continuation-in-part of U.S. patent application Ser. No. 17/883,482, filed on Aug. 8, 2022, which is a continuation-in-part of U.S. patent application Ser. No. 16/711,633, filed on Dec. 12, 2019, which claims priority to U.S. Provisional Patent Application Ser. No. 62/779,702 filed on Dec. 14, 2018, all of which are hereby incorporated by reference in their entirety.

GOVERNMENT SUPPORT

[0002] This invention was made with government support under Grant No. DE-FE-0007395 awarded by the Department of Energy. The government has certain rights in the invention.

TECHNICAL FIELD

[0003] This document relates generally to a new and improved apparatus and method for capturing carbon dioxide (CO₂) from an acid gas stream.

BACKGROUND

[0004] The cleaning of acid gases or sour gas, such as carbon dioxide in particular, from natural gas and in oil refining has been an extensively practiced technology. The industrial removal of carbon dioxide from natural gas dates back to the 1930's.

[0005] In the 21st century, due to the potential impact of anthropogenic carbon dioxide emissions on the climate, post-combustion carbon dioxide capture has gained tremendous attention. While several technologies exist for the removal of acid gases, one of the most commonly employed practices is the use of aqueous amines.

[0006] In this process, an aqueous amine solution is circulated between an absorber/absorption tower and a stripper. The flue gas or acid gas, containing carbon dioxide enters the bottom of the absorber while the aqueous amine absorbent enters the top of the absorber in counter-current flow to the acid gas. As the acid gas and the amine absorbent come into contact in the absorber, the absorbent removes the carbon dioxide from the gas stream. The amine solution, now rich in carbon dioxide, is discharged from the bottom of the absorber and passed through a heat exchanger to improve efficiency before entering the top of the stripper where the amine solution is heated to a higher temperature. The stripper removes the carbon dioxide as a gas from the amine solution. The carbon dioxide is then passed through a condenser and separated from water at a separator. The carbon dioxide is then subjected to downstream processing or storage while the water is returned to the stripper. The carbon dioxide lean amine solution exits the bottom of the stripper and is returned to the absorber by way of the heat exchanger and a chiller.

[0007] The absorber tower is the single biggest capital expenditure in a CO₂ capture process. Estimates put the absorber itself as 30-50% of the entire CO₂ capture capital cost. Therefore, maximizing the effectiveness of the absorber tower, such as achieving maximum liquid-gas interface as described in the prior art, and minimizing the height is an important part of reducing the cost of CO₂ capture. Toward this end, two aspects are both important-

controlling the temperature profile and minimizing the liquid-gas maldistribution inside column.

[0008] During carbon capture, an exothermic chemical absorption reaction occurs in the absorber tower. The reaction is strongest at the top, after the lean solvent enters the column and depends on the local solvent mass to gas mass within the column (L/G). As the reaction proceeds, the temperature increases. As the temperature increases the driving force for CO₂ capture decreases and the reaction slows. Also, as the reaction proceeds the amount of gas (G) decreases, as CO₂ is transferred from the gas phase to the liquid phase. Typically, a temperature bulge occurs in the top section (10-20%) of the column and the column height (30-40% of the column) below this bulge is relatively ineffective.

[0009] We have now found that changing the type of structured packing in the absorber tower allows for different liquid-gas interface areas to be applied at different heights of the column. This allows the temperature and therefore the reaction rate to be controlled and this, in turn, allows advantage to be taken of the entire column height, minimizing the ineffective zone below the temperature bulge. This maximizes the effectiveness of the absorber tower thereby allowing the use of a smaller absorber tower, saving both capital and operating cost.

[0010] Traditionally, liquid and gas is redistributed inside a column with specialized internals that require 5-10 feet of column height. We use a few inches of random packing applied between sections of structured packing as built-in redistributor to cause short sections of higher pressure drop (15-20% of pressure drop of the value produced by section of structure packing above) that work to redistribute the liquid and gas. This will save significant column height and allows for a smaller column to be applied, saving capital cost.

[0011] This document is related to a new and unique apparatus having a discretized packing arrangement, as well as to a related method, that are both adapted to modify this temperature profile, effectively moving the temperature bulge down the absorber thereby resulting in as much as a 5-11% increase in rich loading, depending upon solvent lean loading.

SUMMARY

[0012] In accordance with the purposes and benefits set forth herein, a new and improved apparatus is provided. That apparatus comprises an absorber having a discretized packing arrangement including a first packing segment, a second packing segment and a third packing segment wherein the first packing segment includes a first structured packing having a first specific surface area SA 1, the second packing segment includes a second structured packing having a second specific surface area SA 2 and the third packing segment includes a third structured packing having a third specific surface area SA 3 where SA 1<SA 2<SA 3. One or two or more packing makes up a section separated by pre-installed mechanical gas and liquid distributors.

[0013] More specifically, the second structured packing segment is provided between the first structured packing segment and the third structured packing segment within the absorber. Still more specifically, the first packing segment is provided above the second packing segment and the second packing segment is provided above the third packing segment. Generally, the first and second packing segments will

be combined into one section (the top packing section of absorber), and the third packing segment makes up the bottom section of packing. A set of mechanical gas and liquid distributor is installed between top and bottom packing section.

[0014] In one or more of the many possible embodiments of the absorber, at least one liquid/gas distributor is included in at least one of the first packing segment, the second packing segment and the third packing segment. In one or more of the many possible embodiments of the apparatus, the at least one liquid/gas distributor is a section of random packing.

[0015] For purposes of this document, the terminology “structured packing” refers to a uniform arrangement of packing material or elements. For purposes of this document, the terminology “random packing” refers to randomly fitting material or elements used to increase the surface area over which reactants can interact while minimizing the complexity of the column. For purposes of this document, the packing “segment” refers a group of packing materials with same specification. For purposes of this document, the packing “section” refers a set of packing materials with one or more packing segments.

[0016] Significantly, the random packing used in the absorber is characterized by a random packing unit pressure drop that is greater than any unit pressure drop characteristic of the first structured packing of the first packing segment, the second structured packing of the second packing segment and the third structured packing of the third packing segment.

[0017] In one or more of the many possible embodiments of the apparatus, the first specific surface area SA 1 of the first structured packing is less than $34 \text{ ft}^2/\text{ft}^3$, the second specific surface area SA 2 is between $34 \text{ ft}^2/\text{ft}^3$ and $129 \text{ ft}^2/\text{ft}^3$ and the third specific surface area SA 3 is above $152 \text{ ft}^2/\text{ft}^3$.

[0018] In one or more of the many possible embodiments of the apparatus, the first packing segment includes multiple layers of the first structured packing separated by a first layer of the random packing. In one or more of the many possible embodiments, the multiple layers of the first structured packing have a first thickness T1 of 121.92-182.88 cm and the layer of random packing has a second thickness T2 of 7.62-15.24 cm.

[0019] In one or more of the many possible embodiments of the apparatus, the second packing segment includes multiple layers of the second structured packing having the first thickness T1 separated by a second layer of the random packing having the second thickness T2.

[0020] In one or more of the many possible embodiments of the apparatus, the third packing segment includes multiple layers of the third structured packing having the first thickness T1 separated by a third layer of the random packing having the second thickness T2.

[0021] In one or more of the many possible embodiments of the apparatus, the absorber is adapted to remove carbon dioxide from a flue gas stream using an amine solvent and the apparatus further includes a stripper adapted to remove carbon dioxide from the amine absorbent.

[0022] In one or more of the many possible embodiments of the apparatus, the apparatus may also include a secondary stripper downstream from the stripper. The stripper may include (a) an upper packing section including the third structured packing having the third specific surface area SA 3 and (b) a lower packing section including the first and/or

second structured packing having the first specific surface area SA 1 and the second specific surface area SA 2.

[0023] In one or more of the many possible embodiments of the apparatus, the upper packing section may include multiple layers of the third structured packing separated by a fourth layer of the random packing.

[0024] In one or more of the many possible embodiments of the apparatus, the lower packing section includes multiple layers of the first and/or second structured packing separated by a fifth layer of the random packing.

[0025] In one or more of the many possible embodiments of the apparatus, the secondary stripper includes (a) a top packing section including the second structured packing, having the second specific surface area SA 2, and (b) a lower packing section including a third structured packing, having a third specific surface area SA 3.

[0026] In accordance with an additional aspect, a new and improved method of capturing carbon dioxide from an acid gas stream comprises the steps of: (a) providing an absorber tower with a first packing segment, having a first specific surface area SA 1, a second packing segment, having a second specific surface area SA 2, and a third packing segment, having a third specific surface area SA 3, where $SA 1 < SA 2 < SA 3$ to control local liquid-to-gas mass ratios, CO_2 absorption rates and temperatures in order to reduce a packing height required to achieve a desirable carbon loading; (b) subjecting the acid gas stream to a countercurrent flow of a carbon dioxide lean amine solvent in the absorber tower; (c) separately discharging treated acid gas and a carbon dioxide rich amine solvent from the absorber tower and (d) recovering the carbon dioxide from the carbon dioxide rich amine solvent in a stripper and returning carbon dioxide lean amine solvent to the absorber tower.

[0027] In one or more of the many possible embodiments of the method, the method may include positioning the second packing segment between the first packing segment and the third packing segment. In one or more of the many possible embodiments, the method may further include separating the first and second structured packing segments with a first layer of a higher pressure drop random packing and the second and third structured packing segments with a second layer of higher pressure drop random packing whereby the first and second layers of the higher pressure drop random packing act to redistribute the carbon dioxide lean and the carbon dioxide rich CO_2 absorbent in the absorber tower. In at least one possible embodiment, the method includes the step of controlling an exothermic CO_2 absorption reaction rate by controlling a local liquid-to-gas mass ratio inside the absorber tower.

[0028] In accordance with yet another aspect, a method of capturing carbon dioxide from an acid gas stream, comprises: (a) providing an absorber tower with a plurality of packing segments having different specific surface areas at different heights in the absorber tower to control local liquid-to-gas mass ratios, CO_2 absorption rates and temperatures in order to reduce a packing height required to achieve a desirable carbon loading; (b) subjecting the acid gas stream to a countercurrent flow of a carbon dioxide lean CO_2 absorbent in the absorber tower to produce a carbon dioxide rich CO_2 absorbent; (c) separately discharging treated acid gas and the carbon dioxide rich CO_2 absorbent from the absorber tower; and (d) recovering the carbon dioxide from the carbon dioxide rich CO_2 absorbent in a stripper and returning the carbon dioxide lean CO_2 absorbent to the

absorber tower. In some embodiments, the method further includes the step of separating the plurality of structured packing segments with a higher pressure drop random packing which acts to redistribute the carbon dioxide lean and the carbon dioxide rich CO_2 absorbent in the absorber tower.

[0029] In accordance with yet another aspect, a method of capturing carbon dioxide from an acid gas stream, comprises, consists of or consists essentially of: (a) providing an absorber tower with a first packing segment, having a first specific surface area SA 1 of less than $34 \text{ ft}^2/\text{ft}^3$, a second packing segment, having a second specific surface area SA 2 of between $34 \text{ ft}^2/\text{ft}^3$ and $129 \text{ ft}^2/\text{ft}^3$, and a third packing segment, having a third specific surface area SA 3 of greater than $152 \text{ ft}^2/\text{ft}^3$, to control local liquid-to-gas mass ratios, CO_2 absorption rates and temperatures in order to reduce a packing height required to achieve a desirable carbon loading, (b) subjecting the acid gas stream to a countercurrent flow of a carbon dioxide lean CO_2 absorbent in the absorber tower to produce a carbon dioxide rich CO_2 absorbent, (c) separately discharging treated acid gas and the carbon dioxide rich CO_2 absorbent from the absorber tower, and (d) recovering the carbon dioxide from the carbon dioxide rich CO_2 absorbent in a stripper and returning the carbon dioxide lean CO_2 absorbent to the absorber tower.

[0030] The method may further include or comprise positioning the second packing segment between the first packing segment and the third packing segment. The method may further include or comprise separating the first and second structured packing segments with a first layer of a higher pressure drop random packing and the second and third structured packing segments with a second layer of higher pressure drop random packing whereby the first and second layers of the higher pressure drop random packing act to redistribute the carbon dioxide lean and the carbon dioxide rich CO_2 absorbent in the absorber tower.

[0031] In at least some embodiments, the method may further include or comprise controlling an exothermic CO_2 absorption reaction rate by controlling a local liquid-to-gas mass ratio inside the absorber tower. The method may further include or comprise reducing the packing height required to achieve a desirable carbon loading by 10 percent.

[0032] In some embodiments, the method further includes reducing the packing height required to achieve a desirable carbon loading by 15 percent. In some embodiments, the method further includes reducing the packing height required to achieve a desirable carbon loading by 20 percent. In some embodiments, the method further includes reducing the packing height required to achieve a desirable carbon loading by 25 percent. In still other embodiments, the method further includes reducing the packing height required to achieve a desirable carbon loading by 30 percent.

[0033] In the following description, there are shown and described several embodiments of the apparatus and the related method. As it should be realized, the apparatus and method are capable of other, different embodiments and their several details are capable of modification is various, obvious aspects all without departing from the apparatus and method as set forth and described in the following claims. Accordingly, the drawings and descriptions should be regarded as illustrative in nature and not as restrictive.

BRIEF DESCRIPTION OF THE DRAWING FIGURES

[0034] The accompanying drawing figures incorporated herein and forming a part of the specification, illustrate several aspects of the apparatus and method and together with the description serve to explain certain principles thereof.

[0035] FIG. 1 is a schematic illustration of one possible embodiment of the new and improved apparatus including an absorber, a stripper and a secondary stripper.

[0036] FIG. 2 is a detailed schematic view of the absorber illustrating the new and improved discretized packing arrangement that improves the efficiency of the carbon dioxide capture process.

[0037] FIG. 3 is a detailed schematic view of the stripper illustrating the new and improved discretized packing arrangement that improves the efficiency of the carbon dioxide stripping process.

[0038] FIG. 4 is a detailed schematic view of the secondary stripper illustrating the new and improved discretized packing arrangement that improves the efficiency of the carbon dioxide stripping process.

[0039] FIG. 5 is a graph of packing height vs. temperature that illustrates how the discretized packing arrangement described in this document modifies the temperature profile of the absorber column resulting in a 5-11% increase in rich loading, depending upon solvent lean loading.

[0040] FIG. 6 is a graph illustrating a small plot CO_2 absorber temperature profile where O is the top and 14 is the bottom of the absorber tower. A 100% draw-off intercooler is at 9 m. The temperature bulge of more than 20°C . occurs in the top third of the tower rendering the heights from 6 to 9 m ineffective.

[0041] FIG. 7 is a plot illustrating that a higher temperature absorption rate is attainable with an ideal absorber temperature profile and a lower CO_2 absorption rate is attainable with a typical temperature profile, leading to a taller tower requirement to achieve the same CO_2 capture efficiency.

[0042] FIG. 8 is a plot of 0.7 MWe small pilot absorber temperature profiles (model predictions for liquid and gas (solid lines). Experimental data in dots).

[0043] FIG. 9 illustrates the effectiveness of carbon dioxide absorber along column height where YCO_2 is the carbon dioxide gas phase concentration and $dY \cdot \text{CO}_2$ is the driving force between gas and liquid for mass transfer.

[0044] FIG. 10 is a plot illustrating small bench experimental data showing increase in column pressure drop and rich loading with discretized packing.

[0045] Reference will now be made in detail to the illustrated embodiments of the apparatus, examples of which are illustrated in the accompanying drawing figures.

DETAILED DESCRIPTION

[0046] Reference is now made to FIG. 1 generally illustrating at a high level the new and improved apparatus 10. Flue gas from a coal fired power plant (not shown) is delivered to a direct contact cooler 12 (note action arrow A) where the flue gas is contacted with water being circulated by a pump 14. That water is first cooled by heat exchange with a cooling water supply at heat exchanger 16. As a result, the temperature of the flue gas is reduced to 37.78°C .

48.89° C. Further, water vapor is removed from the flue gas. This minimizes the chemicals used in the downstream pretreatment column 18.

[0047] Next, a blower 20 delivers the flue gas to the pretreatment column 18 where the flue gas is treated with an aqueous solution of soda ash (Na_2CO_3) or sodium hydroxide (NaOH) to remove sulfur dioxide (SO_2). More particularly, the pump 22 circulates the caustic solution through a second cooling water supply heat exchanger 24 if needed and the pretreatment column 18. The removal of the sulfur dioxide serves to minimize thermal stable salt formation and degradation of the solvent in the CO_2 capture block.

[0048] The flue gas is then passed through a membrane 26, of a type known in the art, to split the flue gas into two streams. Such a membrane is more fully described in U.S. Pat. No. 9,409,120 (owned by the assignee of the present invention), the full disclosure of which is incorporated herein by reference.

[0049] The first stream 28 is a CO_2 -enriched permeate stream comprising approximately 24% of the total flue gas and containing approximately 25% CO_2 after removal of water vapor at 40° C. The second stream 30 is a CO_2 -lean reject stream (approximately 76% of the total flue gas flow rate) containing approximately 10% CO_2 after water removal at 40° C.

[0050] More particularly, the two streams 28 and 30 are both delivered to the absorber tower or absorber 32. There the flue gas is subjected to a concurrent stream of an CO_2 absorbent or amine solvent of a type known in the art to be suitable for CO_2 capture. Such a CO_2 absorbent includes primary, secondary and tertiary amines including, for example, 1-amino-2-propanol (A2P), 2-amino-2-methyl-1-propanol (AMP), piperidine (PZ), methyldiethanolamine (MDEA) and other compounds, including, for example, anti-oxidant.

[0051] Following CO_2 removal, the treated flue gas is delivered from the top of the absorber 32 to the solvent recovery column 34 where CO_2 absorbent entrained in the treated flue gas is recovered using a countercurrent flow of wash water and an amine nucleation agent circulated through a separator element 36, such as a screen or filter, by a pump 38. As shown, the wash water, amine nucleation agent and entrained amine solvent may be cooled by a cooling water supply in the heat exchanger 40. Wash water, including entrained amine solvent separated from the amine nucleation agent by the separator element 36 is returned to the absorber 32 by means of a return circuit not shown.

[0052] The solvent recovery column 34 and the method for recovering the amine solvent from the treated flue gas are more fully described in U.S. patent application Ser. No. 16/460,229 entitled APPARATUS AND METHOD FOR RECOVERING AN AMINE SOLVENT FROM AN ACID GAS STREAM and filed on Jul. 2, 2019, the full disclosure of which is incorporated herein by reference. As disclosed therein, the amine nucleation agent may comprise an activated carbon having a density of $997 \text{ kg/m}^3 \pm 600 \text{ kg/m}^3$ and a diameter less than 1.0 millimeter. The treated flue gas may then be discharged.

[0053] The now carbon dioxide rich CO_2 -absorbent or amine solvent is discharged from the bottom of the absorber 32 and directed by the pump 42 through the heat exchanger 44 to the stripper 46 where carbon dioxide is stripped from the carbon rich CO_2 -absorbent or amine solvent. The carbon dioxide exits at the top of the stripper 46 and is routed

through a primary heat recovery exchanger 48 before exiting as a CO_2 product stream that may be stored or undergo other chemical processing (not shown). The reboiler 52 functions to recycle carbon dioxide rich CO_2 -absorbent or amine solvent through the stripper 46 to ensure more efficient processing.

[0054] Carbon dioxide semi-lean CO_2 -absorbent or amine solvent exits the bottom of the stripper 46 and is transferred by pump 54 through the heat exchanger 56 to the secondary stripper 50 where even more carbon dioxide is stripped from the carbon dioxide semi-lean CO_2 -absorbent or amine solvent. This additional carbon dioxide exits the top of the secondary stripper 50 and is routed through the secondary heat exchanger 44 before being recycled to the power plant boiler. The now carbon dioxide lean CO_2 -absorbent or amine solvent exits the bottom of the secondary stripper 50 and is returned by the pump 58 through the lean heat recovery exchanger and the solvent polishing exchanger 62 to the absorber 32 where it is used to capture carbon dioxide from the flue gas as previously described. The water evaporator 61 supplies makeup water to amine loop from energy recovered from the heat exchangers 60 and 48 by operation of the pump 63.

[0055] Reference is now made to FIG. 2 which illustrates the absorber 32 in detail. As illustrated, the absorber 32 includes a first or uppermost packing segment 64, a second or intermediate packing segment 66 and a third or lowermost packing segment 68. The first packing segment 64 includes a first structured packing 65 of high capacity and low efficiency having a first specific surface area SA 1. The second packing segment 66 includes a second structured packing 67 of high efficiency and high capacity having a second specific surface area SA 2. The third packing segment 68 includes a third structured packing 69 of low capacity and high efficiency having a third specific surface area SA 3. SA 1 is less than SA 2 and SA 2 is less than SA 3.

[0056] In one possible embodiment of the apparatus 10, the first structured packing has a first specific surface area SA 1 of less than $34 \text{ ft}^2/\text{ft}^3$, the second structured packing has a second specific surface area SA 2 of between $34 \text{ ft}^2/\text{ft}^3$ and $129 \text{ ft}^2/\text{ft}^3$ and the third structured packing has a third specific surface area SA 3 of above $152 \text{ ft}^2/\text{ft}^3$.

[0057] In the illustrated embodiment, a first in-situ "random packing" liquid/gas distributor 70 is provided between the first packing segment 64 and the second packing segment 66. A second in-situ "random packing" liquid/gas distributor 72 is provided at an intermediate point of the second packing segment 66. Two additional in-situ "random packing" liquid/gas distributors 74, 76 are provided at spaced points in the third packing segment 68. Each of the in-situ liquid/gas distributors 70, 72, 74, 76 may comprise a layer of random packing. That random packing is characterized by a first pressure drop per height PD1 that is greater than any pressure drop per height PD2 of the first structured packing 65 in the first packing segment 64, the second structured packing 67 in the second packing segment 66 and the third structured packing 69 in the third segment 68.

[0058] In one possible embodiment of the apparatus 10, the first or upper segment 64 comprises the top 20-30% of the total height of the top packing 64, 66 and 68. The second segment 66 comprises the middle 30-50% of the total height

of the packing **64**, **66** and **68**. The third segment **68** comprises the bottom 30-40% of the total height of the packing **64**, **66** and **68**.

[0059] In one possible embodiment of the apparatus **10**, every 60.96 to 243.84 cm of structured packing is interrupted by 7.62-15.24 cm of in-situ liquid/gas distributor in the form of random packing. This includes the first structured packing **65** used to form the first packing segment **64**, the second structured packing **67** used to form the second packing segment **66** and/or the third structured packing **69** used to form the third packing segment **68**.

[0060] It has been found that due to the low CO₂ absorption driving force in utility flue gas and the highly viscous nature of second generation advanced amine solvents, the low pressure drop structured packing used in the packing sections **64**, **66** and **68** suffers from a lack of macro-mixing/turbulence between the bulk solvent and the gas-liquid interface. This results in localized channel flow and significantly reduces column effectiveness. The application of short sections 7.62-15.24 cm of high pressure drop random packing in the form of in-situ liquid/gas distributors **70**, **72**, **74** and **76** re-adjusts the pressure and redistributes the liquid within the structured packing **65**, **67**, **69**. As a result, the efficiency of the absorber **32** is significantly enhanced.

[0061] In one possible embodiment of the absorber **32**, the first packing segment **64** includes multiple layers of the first structured packing **65** separated by in-situ gas/liquid distributor of random packing. The multiple layers of the first structured packing **65** may have a first thickness T1 of 121.92-182.88 cm and the layer of random packing may have a second thickness T2 of 7.62-15.24 cm. The second packing segment **66** may include multiple layers of the second structured packing **67** having a first thickness T1 of 121.92-182.88 cm separated by the layer of random packing **72** having a thickness T2 of 7.62-15.24 cm. The third packing segment **68** may have multiple layers of the third structured packing **69** having a thickness T1 of 121.92-182.88 cm separated by the layer of random packing **74** or **76** having a thickness T2 of 7.62-15.24 cm.

[0062] Reference is now made to FIG. 3 which schematically illustrates the stripper **46** in detail. In the illustrated embodiment, the stripper **46** includes an upper packing section **78** and a lower packing section **80**. The upper packing section **78** includes the third structured packing **69** having the third specific surface area SA 3 while the lower packing section **80** includes the first or the second structured packing **65** having the first specific surface area SA 1 and second specific surface area SA 2. Both the upper packing section **78** and the lower packing section **80** may include multiple layers of the structured packing **69**, **65** separated by in-situ liquid/gas distributors **82**, **84** in the form of a layer of random packing of the type previously described. Every 60.96 to 243.84 cm of structured packing **69**, **65** may be interrupted by 7.62-15.24 cm of in-situ liquid/gas distributor **82**, **84** in the form of random packing. Once again, the alternating layered arrangement of structured packing and random packing re-adjusts the pressure and redistributes the liquid within a section **78**, **80** of the structured packing thereby increasing stripper efficiency.

[0063] Reference is now made to FIG. 4 which schematically illustrates the secondary stripper **50** in detail. In the illustrated embodiment, the secondary stripper **50** includes a top packing section **86** and a bottom packing section **88**. The top packing section **86** includes the first structured packing

65 having the first specific surface area SA 1 while the lower packing section **80** includes a fourth structured packing **89** having fourth specific surface area SA 4 where SA 4=SA 2 or SA 3. Both the top packing section **86** and the lower packing section **88** may include multiple layers of the structured packing **65**, **89** separated by in-situ liquid/gas distributors **90**, **92** in the form of a layer of random packing of the type previously described. Every 60.96 to 243.84 cm of structured packing **65**, **89** may be interrupted by 7.62-15.24 cm of liquid/gas distributor **90**, **92** in the form of random packing. Once again, the alternating layered arrangement of structured packing and random packing re-adjusts the pressure and redistributes the liquid within a section **86**, **88** of the structured packing thereby increasing secondary stripper efficiency.

[0064] The apparatus **10** described herein is useful in a method of capturing carbon dioxide from an acid gas stream. That method includes a number of step including providing an absorber tower with a plurality of packing segments having different specific surface areas at different heights in the absorber tower to control local liquid-to-gas mass ratios, CO₂ absorption rates and temperatures in order to reduce a packing height required to achieve a desirable carbon loading. The method also includes the step of subjecting the acid gas stream to a countercurrent flow of a carbon dioxide lean CO₂ absorbent in the absorber tower to produce a carbon dioxide rich CO₂ absorbent. Still further, the method includes the steps of separately discharging treated acid gas and the carbon dioxide rich CO₂ absorbent from the absorber tower and recovering the carbon dioxide from the carbon dioxide rich CO₂ absorbent in a stripper and returning the carbon dioxide lean CO₂ absorbent to the absorber tower. The method also includes the step of separating the plurality of structured packing segments with a higher pressure drop random packing which acts to redistribute the carbon dioxide lean and the carbon dioxide rich CO₂ absorbent in the absorber tower.

[0065] Controlling the exothermic CO₂ absorption reaction rate is accomplished by controlling the local liquid-to-gas mass ratio inside the absorber tower. The reaction is strongest at the top, after the lean solvent enters the tower. As the reaction proceeds, the temperature increases. As the temperature increases the driving force for CO₂ capture decreases and the reaction slows. Also, as the reaction proceeds the amount of gas (G) decreases, as CO₂ is transferred from the gas phase to the liquid phase. Typically, a temperature bulge occurs in the top section of the absorber tower and the tower height below this bulge is relatively ineffective, as illustrated in FIG. 6. The effect of the temperature bulge on the CO₂ absorption rate is illustrated in FIG. 7.

[0066] Changing the type of structured packing allows for different liquid-gas interface areas to be applied at different heights of the absorber tower. This allows the temperature and therefore the reaction rate to be controlled, which, in turn, allows advantage to be taken of the entire tower height, minimizing the ineffective zone below the temperature bulge. The magnitude of the bulge can be reduced and the temperature profile can be optimized in the CO₂ absorber tower, as illustrated in FIG. 5, which is a graph of packing height versus temperature. This graph illustrates how the discretized packing arrangement, including the alternating layers of structured packing and random packing as described above, modify the temperature profile of the

absorber column resulting in a 5-11% increase in CO₂ rich loading, depending upon solvent lean loading. This maximizes the effectiveness of the CO₂ absorber tower and allows for a smaller tower to be applied, saving both capital and operating cost.

[0067] Traditionally, liquid and gas is redistributed inside an absorber tower with specialized internals every 5-10 ft of structure packing height. Significantly, these specialized internals require several feet of additional column height greatly increasing the overall height of the absorber tower. Instead, we use a few inches of random packing applied between sections of structured packing to cause short sections of higher gas pressure drop that work to redistribute the liquid and gas. Keeping the CO₂ absorber tower or column short saves both capital and operating costs.

EXPERIMENTAL

[0068] Absorber and Its Temperature Bulge: Generally, the diameter of an absorption column is determined by the highest flooding point at a given liquid/gas (L/G) ratio, which typically occurs at the temperature bulge, in the top packed section, 10-15% of packing height from the top as shown in FIG. 8. These temperature profiles are confirmed at the UKy-CAER 0.7 MWe small pilot CCS, when L/G~3.2, which is typical for advanced solvents. A temperature rise of ~50° F. is observed 10 feet below the lean amine feed. The temperature bulge represents an imbalance in the CO₂ absorption rate and the exothermic reaction heat rejection. FIG. 9 (left) shows that the result of this heat imbalance is that 20% of packing does the majority of CO₂ capture and the function of the rest of the packing, which is 80% of a tall column, is only to enhance the rich C/N at the outlet of the absorber from 0.47 to 0.52 due to the low absorption rate near the CO₂ driving force pinch point (right).

[0069] In attempting to flatten the temperature profile, increase the effectiveness of column and lower the capital cost, Discretized Packing Arrangement has been developed based on 0.7 M We small pilot experimental results and verified with Aspen Plus® process simulation. The upper section will contain high capacity, low efficiency packing to reduce the CO₂ absorption (less heat released) and reduce channel flow resulting from lower viscosity and surface tension. The middle section will contain high efficiency, high capacity packing to boost the flue gas temperature prior to entering upper section. The bottom section will contain low capacity, high efficiency packing to counter potential channel flow caused by higher viscosity and surface tension. Experimental results CAER bench post-combustion CO₂ capture facility show that the temperature profile and rich loading can be tuned with different packing configurations.

[0070] FIG. 5 shows a modified temperature profile with the bulge moving down from the top of the column with drift cooling observed at the bottom of the column, indicative of the solvent having reached the maximum loading before the end of the column. FIG. 10 shows a 5 and 11% increase in rich loading with a discretized packing configuration, that could effectively reduce the packing height in the range 10-30% of total packing height depended on the solvent properties.

[0071] Each of the following terms written in singular grammatical form: “a”, “an”, and “the”, as used herein, means “at least one”, or “one or more”. Use of the phrase “One or more” herein does not alter this intended meaning of “a”, “an”, or “the”. Accordingly, the terms “a”, “an”, and

“the”, as used herein, may also refer to, and encompass, a plurality of the stated entity or object, unless otherwise specifically defined or stated herein, or, unless the context clearly dictates otherwise. For example, the phrase: “a stripper”, as used herein, may also refer to, and encompass, a plurality of strippers.

[0072] Each of the following terms: “includes”, “including”, “has”, “having”, “comprises”, and “comprising”, and, their linguistic/grammatical variants, derivatives, or/and conjugates, as used herein, means “including, but not limited to”, and is to be taken as specifying the stated component(s), feature(s), characteristic(s), parameter(s), integer(s), or step(s), and does not preclude addition of one or more additional component(s), feature(s), characteristic(s), parameter(s), integer(s), step(s), or groups thereof.

[0073] The phrase “consisting of”, as used herein, is closed-ended and excludes any element, step, or ingredient not specifically mentioned. The phrase “consisting essentially of”, as used herein, is a semi-closed term indicating that an item is limited to the components specified and those that do not materially affect the basic and novel characteristic(s) of what is specified.

[0074] Terms of approximation, such as the terms about, substantially, approximately, etc., as used herein, refers to ±10% of the stated numerical value.

[0075] The foregoing has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the embodiments to the precise form disclosed. Obvious modifications and variations are possible in light of the above teachings. All such modifications and variations are within the scope of the appended claims when interpreted in accordance with the breadth to which they are fairly, legally and equitably entitled.

What is claimed:

1. A method of capturing carbon dioxide from an acid gas stream, comprising:

providing an absorber tower with a first packing segment, having a first specific surface area SA 1 of less than 34 ft²/ft³, a second packing segment, having a second specific surface area SA 2 of between 34 ft²/ft³ and 129 ft²/ft³, and a third packing segment, having a third specific surface area SA 3 of greater than 152 ft²/ft³, to control local liquid-to-gas mass ratios, CO₂ absorption rates and temperatures in order to reduce a packing height required to achieve a desirable carbon loading; subjecting the acid gas stream to a countercurrent flow of a carbon dioxide lean CO₂ absorbent in the absorber tower to produce a carbon dioxide rich CO₂ absorbent; separately discharging treated acid gas and the carbon dioxide rich CO₂ absorbent from the absorber tower; and

recovering the carbon dioxide from the carbon dioxide rich CO₂ absorbent in a stripper and returning the carbon dioxide lean CO₂ absorbent to the absorber tower.

2. The method of claim 1, including positioning the second packing segment between the first packing segment and the third packing segment.

3. The method of claim 2, further including separating the first and second structured packing segments with a first layer of a higher pressure drop random packing and the second and third structured packing segments with a second layer of higher pressure drop random packing whereby the first and second layers of the higher pressure drop random

packing act to redistribute the carbon dioxide lean and the carbon dioxide rich CO₂ absorbent in the absorber tower.

4. The method of claim 1, further including controlling an exothermic CO₂ absorption reaction rate by controlling a local liquid-to-gas mass ratio inside the absorber tower.

5. The method of claim 1, further including reducing the packing height required to achieve a desirable carbon loading by 10 percent.

6. The method of claim 1, further including reducing the packing height required to achieve a desirable carbon loading by 15 percent.

7. The method of claim 1, further including reducing the packing height required to achieve a desirable carbon loading by 20 percent.

8. The method of claim 1, further including reducing the packing height required to achieve a desirable carbon loading by 25 percent.

9. The method of claim 1, further including reducing the packing height required to achieve a desirable carbon loading by 30 percent.

10. A method of capturing carbon dioxide from an acid gas stream, comprising:

providing an absorber tower with a plurality of packing segments having different specific surface areas at different heights in the absorber tower to control local liquid-to-gas mass ratios, CO₂ absorption rates and temperatures in order to reduce a packing height required to achieve a desirable carbon loading;

subjecting the acid gas stream to a countercurrent flow of a carbon dioxide lean CO₂ absorbent in the absorber tower to produce a carbon dioxide rich CO₂ absorbent;

separately discharging treated acid gas and the carbon dioxide rich CO₂ absorbent from the absorber tower; and

recovering the carbon dioxide from the carbon dioxide rich CO₂ absorbent in a stripper and returning the carbon dioxide lean CO₂ absorbent to the absorber tower.

11. The method of claim 10, further including separating the plurality of structured packing segments with a higher pressure drop random packing which acts to redistribute the carbon dioxide lean and the carbon dioxide rich CO₂ absorbent in the absorber tower.

12. The method of claim 11, including providing (a) an upper packing segment, having a first specific surface area SA 1 of less than 34 ft²/ft³, (b) an intermediate packing segment, having a second specific surface area SA 2 of between 34 ft²/ft³ and 129 ft²/ft³, and (c) a lower packing segment, having a third specific surface area SA 3 of greater than 152 ft²/ft³, to control local liquid-to-gas mass ratios, CO₂ absorption rates and temperatures in order to reduce a packing height required to achieve a desirable carbon loading.

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